



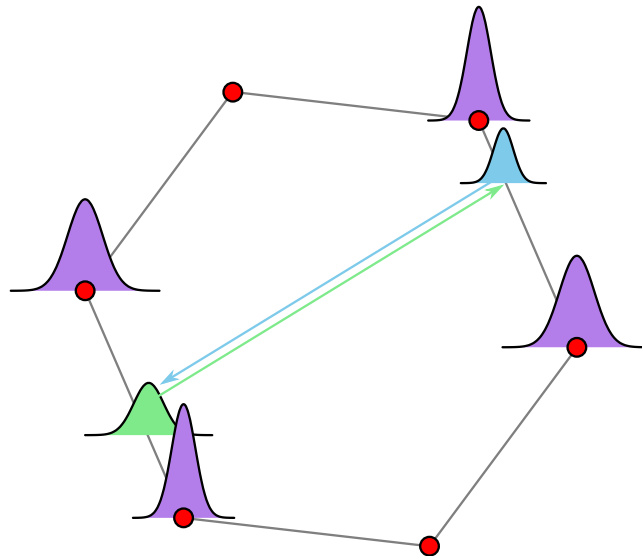
THE UNIVERSITY OF
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FINAL REPORT

The Art of Scientific Computing: Complex Systems

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Chapter 1

Introduction

This chapter introduces the background of self-organized criticality (SOC) and the Bak–Tang–Wiesenfeld (BTW) sandpile model. It also explains the theoretical importance of the Abelian property and the main aim of this project.

1.1 Complex Systems and Self-Organized Criticality

In many physical systems, the size of an effect is often related to the size of its cause. For example, there may only be a local impact when a small meteorite hits the Earth, but a much larger object may influence the whole orbital system. However, in complex system, this simple relationship does not always hold. In some complex systems, a small local perturbation can lead to a large-scale event [4].

A complex system usually contains many components which could interact with each other. Since these components affect each other, the global behaviour of the system is difficult to predict only from the local rules [4]. And some systems can produce events with very different sizes. For example, a small change of stress may cause a large earthquake; a small ignition source may cause a large forest fire; a small fluctuation in a financial market may also spread and lead to a wider market crash.

Bak, Tang, and Wiesenfeld introduced the idea of self-organized criticality (SOC) in 1987 [2]. The SOC describes systems which can automatically evolve to a critical state, without tuning external parameters. In the critical state, the system can produce both small and large events. The system also often shows scale-invariant behaviour [7].

Scale invariance means that, at different scales, the statistical behaviour of the system is similar [7]. And scale invariance is closely related to power-law behaviour [1]. This means that small events happen very frequently, while large events happen less frequently but they are still possible. For example, this type of behaviour is commonly observed in earthquakes, forest fires, and financial market fluctuations [1] [7].

Mathematically, a power-law distribution can arise from scale-type variable transformations [6]. If the probability measure is preserved under a change of variable,

$$P_X(x) dx = P_Y(y) dy,$$

then the transformed distribution depends on the Jacobian of the transformation. For a scale-type transformation such as

$$y = x^{-1/\alpha}, \quad \alpha > 0,$$

the transformed probability density becomes

$$P_Y(y) = P_X(x) \left| \frac{dx}{dy} \right|.$$

If $P_X(x)$ is approximately constant near the origin, this gives

$$P_Y(y) \propto y^{-(1+\alpha)},$$

which is a power-law distribution.

This shows that scale-related transformations can naturally produce power-law behaviour. Therefore, power-law distributions are a natural description for systems with scale-invariant behaviour (**Ans. to Q1**).

For studying SOC, the Bak–Tang–Wiesenfeld (BTW) sandpile model is a simple model to begin with. Although its rules are simple, the sandpile model can produce complex avalanche behaviour. By adding grains of sand slowly and applying local toppling rules, the model can reach a critical state and show statistics with power-law [2]. Therefore, the BTW model is useful for us to study complex systems, self-organized criticality, and non-equilibrium statistical behaviour.

1.2 The Bak–Tang–Wiesenfeld (BTW) sandpile model

The Bak–Tang–Wiesenfeld (BTW) sandpile model simulates a real sandpile on a finite two-dimensional lattice. Although the real sandpile is not created exactly on a the grid, this simplified model keeps the main mechanisms of the system, which include slow input, local instability, chain reactions, and boundary loss [2].

In the BTW model, the sandpile is performed on an $N \times M$ grid. Each site (i, j) has a height $h(i, j)$ as an integer, which represents the number of sand grains at that site. At each time step, one grain of sand is added to one site. This process represents a slow and local external driving force [8].

When the height of a site reaches or exceeds the given stability threshold, the site becomes unstable. In the standard two-dimensional BTW model, the threshold is usually denoted as 4. If a site satisfies:

$$h(i, j) \geq 4,$$

then this site topples.

During a toppling process, the site loses 4 grains of sand, and these grains are assigned to its four orthogonal neighbours: up, down, left, and right [2]. This rule can be mathematically written as

$$h(i, j) \leftarrow h(i, j) - 4,$$

$$h(i \pm 1, j) \leftarrow h(i \pm 1, j) + 1,$$

$$h(i, j \pm 1) \leftarrow h(i, j \pm 1) + 1.$$

If a toppling occurs at the grid boundary, some grains may be sent outside the lattice, and these grains will be removed from the system and counted as boundary loss. The boundary loss is important, because it prevents the total amount of sand on the grid from increasing without limit. With continuous sand addition and boundary loss, the system can eventually reach a statistically stable state [2] [8].

The toppling of one site may cause an unstable condition for its neighbours, and these kind of sites may topple and create further unstable sites. Then a chain reaction of topplings happens, and this event caused by adding one grain of sand is called an avalanche [8]. The behaviour of avalanches is the main object studied in this project, because it shows how a small local input can create events with different sizes.

In the simulation in this study, four avalanche properties are recorded: topples, area, loss, and length. The value of topples is the total number of toppling events in one avalanche; The value of area is the number of unique sites that topple at least once. The value of loss is the number of grains that leave the lattice through the boundary. The value of length measures the spatial extent of the avalanche, in this study, length is defined as the avalanche radius.

By implementing above sand addition, toppling, and avalanche simulation, we can study the behaviour of the system, the distributions of avalanche properties, the correlations between variables.

1.3 Abelian Property and Theoretical Importance

Besides studying avalanche through simulation, the BTW sandpile model also has an important theoretical property, which is called the Abelian property. This property means that, during one avalanche,

the order of topplings does not influence the final relaxed configuration. If there are several unstable sites at the same time, these sites can be toppled in different orders. Dhar states that toppling site i before site i' gives the same configuration as toppling site i' before site i , so the toppling operations commute [3]. Therefore, when all necessary topplings are completed, the final stable sandpile configuration does not depend on the chosen toppling order.

This property is important for the simulation. In one avalanche, there may be many topplings, and there may be more than one unstable site at the same time. If the model did not have the Abelian property, the code would need to control the exact order of topplings very carefully. This would make the simulation more difficult to implement. However, because the BTW model has the Abelian property, the avalanche can be processed using a simpler algorithm.

The Abelian property is also important from a theoretical aspect. It shows that the BTW model has a clear mathematical structure, even though its local rules are simple. In the project document, this property is introduced by using the toppling matrix, the toppling operator, and the stabilization operator. The toppling matrix describes how the height variables are updated when a site topples [3]. In our project notation, the toppling operator represents this local update from one configuration to another. The stabilization operator then represents the full relaxation process, where an unstable configuration is turned into a stable configuration through a sequence of topplings [3].

Therefore, the abelian property directly supports the code design and the reliability of the simulation results in this project. This allows the simulation to record avalanche properties such as topples, area, loss, and length in a consistent way.

1.4 Aim of The Project

The main aim of this project is to implement a Bak–Tang–Wiesenfeld (BTW) sandpile model, and use it to study self-organized criticality and the related statistics on a finite two-dimensional lattice. At the end, we will extend the model to investigate different system behaviours.

Chapter 2

Model Description

In this project, we use the BTW Sandpile Model to study self-organized criticality (SOC). The model uses a finite two-dimensional lattice to perform a sandpile system, and each lattice site corresponds to a local sand pile. In this chapter, we will introduce a detailed mathematical definition for the sandpile model, including the local rules.

2.1 The BTW Sandpile Model

The two-dimensional lattice is defined as:

$$L = \{1, \dots, N\} \times \{1, \dots, M\}$$

where:

- N is the number of rows;
- M is the number of columns;
- $(i, j) \in L$ represents a lattice site.

At time step t , the number of sand grains at site (i, j) is given by:

$$p_t(i, j) \in \mathbb{N}_0.$$

The full sandpile configuration at time t is written as:

$$P_t = \{p_t(i, j)\}_{(i, j) \in L}.$$

In this model, t represents the external sand addition step, rather than the number of internal topplings during an avalanche. At each time step, one grain of sand is added to the system, while the internal toppling process is considered as instantaneous [5].

2.2 Toppling Dynamics and Avalanche Process

At each time step t , we randomly select a lattice site $(i_t, j_t) \in L$, and then add one grain of sand on it:

$$p_t(i_t, j_t) \leftarrow p_t(i_t, j_t) + 1.$$

The model uses a fixed stability threshold:

$$k = 4.$$

A site becomes unstable when:

$$p_t(i, j) \geq 4.$$

When a site becomes unstable, a toppling occurs on it. During one toppling process, the site loses four grains of sand, and one grain is transferred to each of its four nearest neighbours [5]:

$$p_t(i, j) \leftarrow p_t(i, j) - 4,$$

$$p_t(i \pm 1, j) \leftarrow p_t(i \pm 1, j) + 1,$$

$$p_t(i, j \pm 1) \leftarrow p_t(i, j \pm 1) + 1.$$

Note that we only consider the nearest neighbours, without considering diagonal neighbours [5]. The nearest neighbour set is defined as:

$$\mathcal{N}(i, j) = \{(i + 1, j), (i - 1, j), (i, j + 1), (i, j - 1)\}$$

In this project, we use the open boundary condition. If a toppling happens at the boundary of the lattice, some grains may be removed outside the lattice and system.

A single grain addition may cause a chain reaction of topplings. When one unstable site topples, its neighbouring sites may also become unstable and then begin toppling. This process is called an avalanche. In this project, an avalanche is defined as the full sequence of topplings caused by one grain addition until the system becomes stable [5], which is given by:

$$p_t(i, j) < k, \quad \forall (i, j) \in L.$$

2.3 Avalanche Properties

In order to study the avalanche behaviour, this project considers four avalanche properties: topples, area, loss, and radius [5].

We denote the avalanche topples as s , which is the total number of topplings during one avalanche. Note that if one site topples multiple times during the same avalanche, then every toppling is counted in s ; We denote the avalanche area as a , which is the number of unique lattice sites that topple during one avalanche; We denote the avalanche loss as l , which is the total number of sand grains that leave the lattice during one avalanche.

We denote the avalanche length as r , which is defined as the spatial radius:

$$r = \max_{x \in A} \|x - x_0\|_2$$

where:

- x_0 is the dropping site of the avalanche;
- A is the set of all toppled lattice sites;
- $\|\cdot\|_2$ represents the Euclidean distance.

This length measures the maximum Euclidean distance between the dropping site and all toppled sites, which represents how far the avalanche spreads on grid.

After each avalanche, the simulation records the corresponding values of avalanche properties as (s, a, l, r) , which is used for the following statistical analysis.

2.4 Example of an Avalanche Process

In order to clearly illustrate the toppling dynamics and demonstrate how to calculate avalanche properties in the BTW sandpile model, we use a 3×3 lattice avalanche example:

$$\begin{bmatrix} 1 & 3 & 3 \\ 2 & 4 & 2 \\ 2 & 1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 4 & 3 \\ 3 & 0 & 3 \\ 2 & 2 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 0 & 4 \\ 3 & 1 & 3 \\ 2 & 2 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 1 & 0 \\ 3 & 1 & 4 \\ 2 & 2 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 1 & 1 \\ 3 & 2 & 0 \\ 2 & 2 & 2 \end{bmatrix}$$

Here we assume the dropping site is $(2, 2)$. The avalanche process of this example is derived by implementing the definition and rules as shown above. Then we calculate the avalanche properties by following the definition.

First, the avalanche topples s denote the total number of toppling events during the avalanche. In this example, four topplings happen at the following lattice sites:

$$(2, 2), \quad (1, 2), \quad (1, 3), \quad (2, 3),$$

then,

$$s = 4.$$

Next, the avalanche area a denotes the number of unique lattice sites that topple during the avalanche. Since all four toppled sites are different:

$$a = 4.$$

The avalanche loss l denotes the number of sand grains that leave the system through the lattice boundary. In this example:

- the central site $(2, 2)$ is an interior site and produces no loss;
- the upper boundary site $(1, 2)$ has one missing neighbour and produces one grain loss;
- the upper-right corner site $(1, 3)$ has two missing neighbours and produces two grain losses;
- the right boundary site $(2, 3)$ has one missing neighbour and produces one grain loss.

Then, the total loss is:

$$l = 0 + 1 + 2 + 1 = 4$$

Finally, the avalanche radius r is defined as the maximum Euclidean distance between the dropping site and all toppled sites:

$$r = \max_{x \in A} \|x - x_0\|_2$$

where the dropping site is $x_0 = (2, 2)$, and the set of toppled sites is $A = \{(2, 2), (1, 2), (1, 3), (2, 3)\}$. Then the corresponding Euclidean distances are $0, 1, 1, \sqrt{2}$, and:

$$r = \sqrt{2}$$

Hence, the avalanche observables for this example are:

$$s = 4, \quad a = 4, \quad l = 4, \quad r = \sqrt{2}.$$

Chapter 3

Model implementation

3.1 Implementation

3.2 Results and discussion

Chapter 4

Extension

If time allows, is to extend the model by changing how sand is added.

Appendix A

Program flowchart

A flowchart is a conceptual aid which highlights input and output that a program should have, as well as the consequences of any decisions that it must make. Sometimes, these will include *pseudocode*, which mimics some of the more common coding structures in a universal way. (For instance, you might say ‘loop over all array addresses’.)

This particular flowchart indicates a program that reads three numbers from the terminal, and then produces the largest of the numbers back to the terminal. The colour scheme is not mandatory, but helps the reader to know what is going on at a glance.

This flowchart was drawn with the free ‘Lucidcharts’ extension to Google Docs, from which the result was exported as an `svg` file, and any unnecessary features were then cropped out with Inkscape. Choose any method you wish to create your own flowchart!

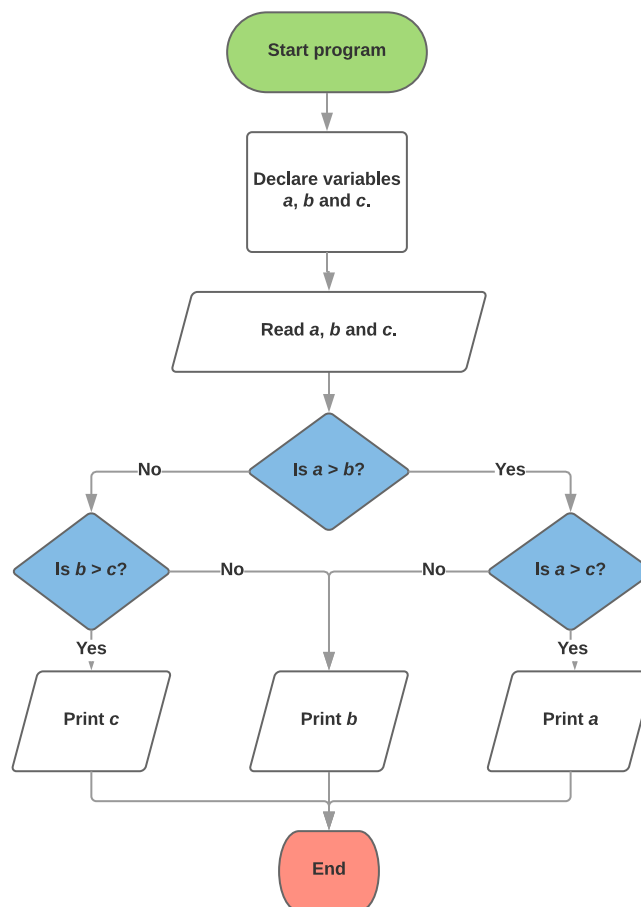


Figure A.1: A flowchart for the determination of the largest of three user-selected numbers.

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